
LECTURE NOTES

Exam preparations

AI505 - Optimization

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Bracketing

For $a < b < c$

there exists a minimum in interval $[a, c]$ if

$$f(a) < f(b) \wedge f(c) < f(b)$$

Fibonacci Search

When we know how many evaluations we have available

Places evaluations such that one evaluation always carries over to the next iteration.

Golden Ratio Search

If one were to use fibonacci with ∞ function evaluations. it would converge to a avg distance of ϕ

Golden Ratio just uses ϕ

Quadratic fit search

Iteratively fits a quadratic and uses that minimum

Lipschitz continuous

Upper bound on the derivative

For some constant L ,

$$|\nabla f| \leq L$$

Bisection method

Find roots since if $\text{sign}(\nabla f(a)) \neq \text{sign}(\nabla f(b))$ then $[a, b]$ guaranteed to contain a root.

So cut the midpoint and choose the side where this property still holds

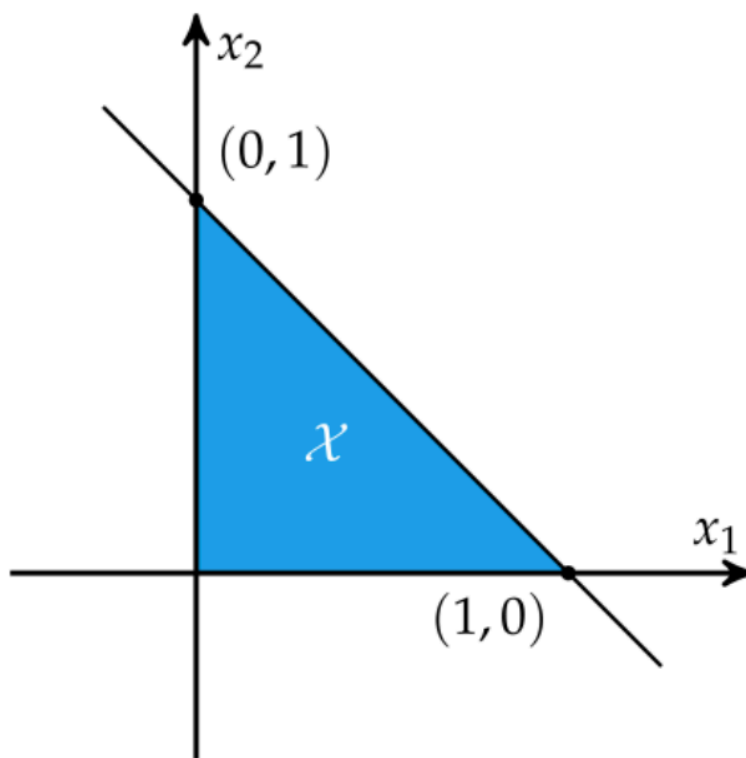
A simple minimization problem

Defining a minimization problem such that x lies within a feasible set of solutions

$$\begin{aligned} \min_x & f(x) \\ \text{s.t.} & x \in \mathcal{X} \end{aligned}$$

Example

$$\begin{array}{ll}\min_{x_1, x_2} & f(x_1, x_2) \\ \text{s.t.} & x_1 \geq 0 \\ & x_2 \geq 0 \\ & x_1 + x_2 \leq 1\end{array}$$



Taylor expansion

Taylor expansion

Let a be a fixed point. The Taylor expansion approximates $f(x)$ near a using only information about f at a .

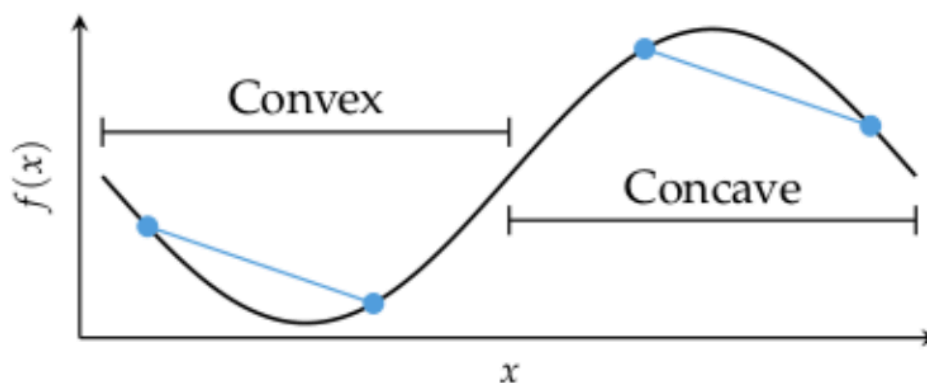
$$f(x) = f(a) + \nabla f(a)^T (x - a) + \frac{1}{2} (x - a)^T \nabla^2 f(a) (x - a)$$

$$f(a + h) = f(a) + \frac{\nabla f(a)}{1!} h + \frac{\nabla^2 f(a)}{2!} h^2 + O(\|h\|^3)$$

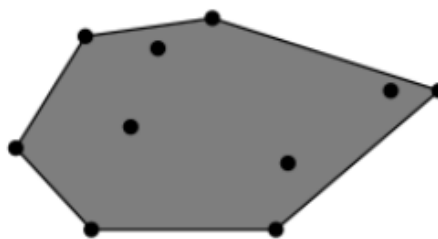
Convexity

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$



For a convex hull:



the convex hull of X

$$\text{conv}(X) = \left\{ \sum_i \lambda_i x_i \mid x_i \in X, \lambda_i \geq 0, \sum_i \lambda_i = 1 \right\}$$

Directional derivative

$$\nabla_s f(x) = \nabla f(x)^T s = \frac{\partial f(x)}{\partial x_1} s_1 + \frac{\partial f(x)}{\partial x_2} s_2 + \dots + \frac{\partial f(x)}{\partial x_n} s_n$$

Positive Definiteness

Positive definiteness

A square, symmetric matrix A is **positive definite** if $x^T A x$ is positive for all points other than the origin

$$A \succ 0 = x^T A x > 0 \quad \forall x \in \mathbb{R} \setminus \{0\}$$

Positive semidefiniteness

A square, symmetric matrix A is **positive semidefinite** if $x^T A x$ is always nonnegative

$$A \succeq 0 = x^T A x \geq 0 \quad \forall x \in \mathbb{R}$$

Descent Direction Iteration

$$x_{k+1} = x_k + \alpha_k d_k \quad \text{where } d = -B_k^{-1} \nabla f(x_k)$$

Note that this is because in Newton $B_k = \nabla^2 f(x_k)$

And for 1st order methods like GD $B_k = I$

Line Search

$$\alpha^* = \min_{\alpha \geq 0} f(x_k + \alpha d_k)$$

Backtracking

$$\alpha_0 = 1$$

Then decrease alpha such that

$$f_{k+1} \leq f_k + \beta \alpha \nabla_{d_k} f_k \quad \text{where } \beta \in [0, 1]$$

Wolfe conditions

1. First Wolfe Condition: Sufficient Decrease (Armijo condition)

$$f(x_{k+1}) \leq f(x_k) + \beta \alpha \nabla f(x_k)^T \nabla_{d_k} f(x_k)$$

1. Second Wolfe Condition: Curvature Condition

$$\nabla_{d_k} f(x_{k+1}) \geq \sigma \nabla_{d_k} f(x_k)$$

Strong wolfe

Modify Curvature Condition such that

$$|\nabla_{d_k} f(x_{k+1})| \geq \sigma |\nabla_{d_k} f(x_k)|$$

Strong Backtracking

- Bracketing-phase: golden ratio
- Zoom phase: bisection or some interpolation (like quadratic fit)

Conjugate Gradient

Using line search results in

$$\nabla f_{k+1}^T \nabla f_k = 0 \quad (\nabla f(x_{k+1}) \perp \nabla f(x_k))$$

$$d_i^T H d_j = 0, \quad \forall i \neq j \text{ Vectors are not orthogonal.}$$

Where $H \succeq 0$

Generally $d_0 = -\nabla f(x_0)$ and,

$$d_{k+1} = -\nabla f_{k+1} + \beta_k \nabla f_k$$

Where

$$\beta_k^{\text{Fletcher-Reaves}} = \frac{\nabla f_{k+1}^T \nabla f_{k+1}}{\nabla f_k^T \nabla f_k}$$
$$\beta_k^{\text{Polak-Ribière}} = \frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{\nabla f_k^T \nabla f_k}$$

Rate of convergence

We need to make sure of method convergence.

by

Theorem: Zoutendijk condition

Consider any iteration of the form $x_{k+1} \leftarrow x_k + \alpha_k d_k$, where d_k is a descent direction and α_k satisfies the Wolfe conditions. Suppose that f is bounded below in $\mathbb{R}^n$ and that f is continuously differentiable in an open set N containing the level set $L = \{x : f(x) \leq f(x_0)\}$, where x_0 is the starting point of the iteration. Assume also that the gradient ∇f is Lipschitz continuous on N , that is, there exists a constant $L > 0$ such that:

$$|f(x) - f(y)| \leq L|x - y|, \forall x, y \in N$$

Then:

$$\sum_{k \geq 0} \cos^2 \theta_k \|\nabla f_k\|^2 < \infty$$

This means that $\cos^2 \theta_k \|\nabla f_k\|^2 \rightarrow 0$

Since $\cos \theta_k$ can be bounded by a constant $\frac{1}{M}$ we know that

$$\|\nabla f_k\|^2 \rightarrow 0$$

For a sequence of x in $\mathbb{R}^n$

$$\text{Q-linear} \approx \text{R-linear} < \text{Q-superlinear} < \text{Q-quadratic}$$

Q-linear (quotient-linear)

There exists $r \in (0, 1)$ such that

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \leq r \quad \forall k \gg 1$$

Q-superlinear (quotient-superlinear)

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0$$

Q-quadratic (quotient-superlinear)

For some constant $M > 0$

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|^2} \leq M \quad \forall k \gg 1$$

R-linear (root-linear)

For a sequence of nonnegative scalars $\{v_k\}$ where $v_k \rightarrow 0$

$$\|x_{k+1} - x^*\| \leq v_k \quad \forall k \gg 1$$

Trust Region Methods

$$\begin{aligned} & \min_{x'} \hat{f}(x') \\ & s.t. \quad \|x - x'\| \leq \delta \end{aligned}$$

δ can be expanded or contracted by η

$$\eta = \frac{\text{real improvement}}{\text{predicted improvement}} = \frac{f(x) - f(x')}{f(x) - \hat{f}(x')}$$

Termination Conditions

- Maximum Iterations: $k > k_{\max}$
- Absolute Improvement: $f(x_k) - f(x_{k+1}) < \varepsilon_a$
- Relative Improvement: $f(x_k) - f(x_{k+1}) < \varepsilon_r |f(x_k)|$
- Gradient Magnitude: $\|\nabla f(x_{k+1})\| < \varepsilon_g$

Steepest Descent

$$d_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$$

Conjugate gradient method

In generating its set of conjugate vectors, it can compute a new vector d_k by using only the previous vector d_{k-1} .

$$d_k = -\nabla f(x_k) + \beta_k d_{k-1}$$

Where

$$\beta_k = \frac{\nabla f(x_k)^T A d_{k-1}}{d_{k-1}^T A d_{k-1}}$$

Momentum

$$\begin{aligned} v_{k+1} &= \beta v_k - \alpha \nabla f(x_k) \\ x_{k+1} &= x_k + v_{k+1} \end{aligned}$$

Newton method

Multivariate update rule

$$x_{k+1} = x_k - H_k^{-1} \nabla f(x_k)$$

Convergence

When $\nabla^2 f(x^*) \succ 0$ and $\nabla f(x^*) = 0$ (locally bowl shaped) $\implies \min_x = x^*$

When f is strong convex, then $\nabla^2 f(x) \succ 0, \quad \forall x$

Secant method

Approximate $f''(x_k) \approx \frac{f'(x_k) - f'(x_{k-1})}{x_k - x_{k-1}}$ So

$$x_{k+1} = x_k - \frac{f'(x_k) - f'(x_{k-1})}{x_k - x_{k-1}} f'(x_k)$$

Quasi-Newton methods

Approximate $Q_k \approx H^{-1}$ iteratively

Derivative free methods

- Cyclic Coordinate Search
 - line search in alternating coordinate directions
- Powell's method
 - Drops already seen directions
- Hooke-Jeeves
 - evaluate $f(x)$ and $f(x \pm \alpha e_i)$ and select best one
 - If no improvements are found shrink α
- Generalized pattern search
 - Hooke-Jeeves but with $n + 1$ directions
- Nelder-Mead
 - Reflection, expansion, contraction, shrinkage.
- DIRECT - Divided rectangles
 - Balances interval size and function evaluation

Stable Step Size

Consider

$$f(x) = \frac{1}{2} x^T Q x - b^T x, \quad \text{where } Q = Q^T \succ 0$$

Then $\nabla f(x) = Qx - b$ One can derive that the step size that stabilizes the iteration

$$0 < \alpha < \frac{2}{\lambda_{\max}(Q)}$$

Noisy Descent

$$x_{k+1} = x_k + \alpha \nabla f(x_k) + \varepsilon_k \quad \text{where } \varepsilon_k \sim \mathcal{N}(0, \sigma_k^2)$$

Stochastic Gradient Descent

$$\nabla f(x) \approx \frac{1}{|B|} \sum_{i \in B} \nabla f_i(x)$$

Mesh Adaptive Direct Search

Similar to generalized pattern search but uses controlled randomness to generate directions

Simulated Annealing

$$x' = x + \varepsilon, \quad \text{where } \varepsilon \in \mathcal{N}(0, T)$$

Metropolis acceptance criterion:

Let $\Delta = f(x') - f(x)$ Then

$$p(x, x') = \begin{cases} 1 & \text{if } \Delta \leq 0 \\ e^{-\Delta/t_k} & \text{if } \Delta > 0 \end{cases}$$

Cross entropy/Max likelihood

Fit a distribution to elite points (lowest function evaluation)

$$\theta^* = \arg \min_{\theta} - \sum_{i=1}^N \log p(x_i | \theta) = \arg \min_{\theta} \prod_{i=1}^N p(x_i | \theta)$$

One can model data and use to sample from and/or make predictions.

Natural Evolution Strategies

Now gradient descent:

$$\theta_{k+1} = \theta_k - \alpha \frac{1}{N} \sum_{i=1}^N f(x_i) \nabla_{\theta} \log p(x_i | \theta)$$

Where $p(\cdot | \theta) = \mathcal{N}(\mu, \sigma^2)$

Similar to CE but all points.

CMA-ES

Same but with covariance so that it can utilize mutivariate data.

$$x \sim \mathcal{N}(\mu, \sigma^2 \Sigma)$$

mean is weighed average of m -elites

CMA-ES uses cumulative vectors p_1 (length) and p_2 (shape), which a decaying memory vector $\in \mathbb{R}^n$ (kinda like rnn's)

COCO benchmarking

Benchmark your method against 24 functions

- n dimensions
- i function (1-24)
- j instance (rotate/shift)

Population-based methods

Optimize a collection of **individuals**

We only keep track of chromosomes (each individual is interpreted as a set of chromosome, commonly vectors $\in \mathbb{R}^n$)

- **Selection**
 - Determine which individuals pass their genetic information on to the next generation
 - **Truncation** selection: truncate the lowest performers
 - **Tournament** selection: selects fittest out of k randomly chosen individuals
 - **Roulette Wheel** selection: individuals are chosen with probability proportional to their fitness
- **Crossover**
 - Combine chromosomes from two parents to create child
 - **Single-point**: swap occur after a single point
 - **Multiple-point**: swap z times
 - **Uniform** crossover: Each chromosome has 50% chance of crossover
- **Mutation**: randomly mutate some individuals to support explorations.
 - Mutate by *mutation rate*
- **Replace**: Choose the new populations using all above
 - **Goal**: find the best population, but avoid duplicates.

Differential Evolution

Choose 3 random distinct individuals a, b and c

Then

$$z = a + w(b - c)$$

$$x'_i \begin{cases} z_i & \text{if } i = j \text{ where } j \sim \mathcal{U}\{1, \dots, n\} \\ x_i & \text{otherwise} \end{cases}$$

Swarm

Now let each individual track,

- position,
- velocity
- Own best position
- Global best position

As particles fly toward the current global best, they pass through different regions of the search space. If one of them stumbles across a better point along the way — that becomes the new global best, and the whole swarm redirects toward it.

Sampling Plans

Choosing how to initialize points before optimization begins. There are several methods.

- try optimizing multiple starting points and keep best result
- if f is expensive, one can sample points from a *surrogate model*

Surrogate model: cheap fake version of f

Full Factorial Design

Sample points uniformly and evenly spaced samples across domain Simple, easy to design, but grows with n^m and can miss local features

Random Sampling

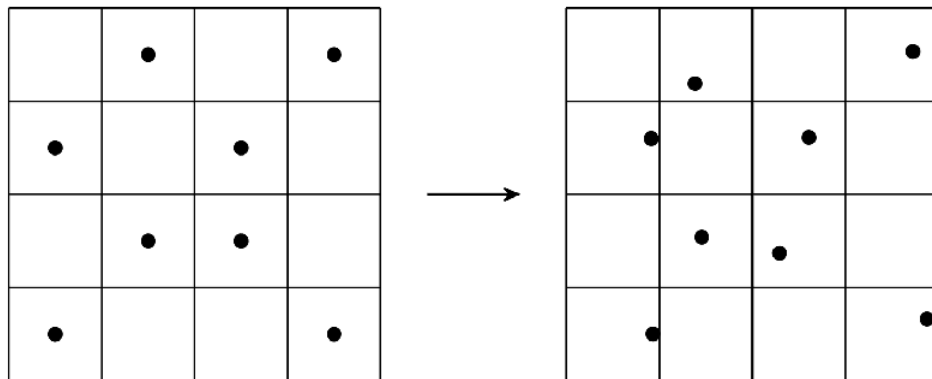
Sample randomly from some distribution over $[a_1, b_1] \times \dots \times [a_n, b_n]$

Uniform Projection Plans

Sample m points from $m \times m$ with the premise that every dimension must be uniformly covered (like N-queens but with rooks).

Stratified Sampling

Fix problem with local accuracy that full factorial has. Place points uniformly random within their square



Space Filling Metrics

SFM: Discrepancy

SFM: Pairwise Distances

SFM: Morris-Mitchell Criterion

Space-Filling Subsets

Quasi-Random Sequences

Lagrangian Function

$$\mathcal{L}(x, \mu, \lambda) = \sum_i \mu_i g_i(x) + \sum_j \lambda_j h_j(x)$$

KKTC (Karush-Kuhn-Tucker conditions)

For any x^*

1. Primal feasibility: $g(x^*) \leq 0, h(x^*) = 0$
2. Dual feasibility: $\mu \geq 0$
3. Complementary slackness: $\mu \cdot g(x^*) = 0$
4. Stationarity: $\nabla f(x^*) + \mu \cdot \nabla g(x^*) + \lambda \cdot \nabla h(x^*) = 0$

Primal form

$$\min_x \max_{\mu \geq 0, \lambda} \mathcal{L}(x, \mu, \lambda)$$

Dual form

$$\max_{\mu \geq 0, \lambda} \min_x \mathcal{L}(x, \mu, \lambda)$$

Penalty methods

Examples:

- Count penalty

$$p_{\text{count}}(x) = \sum_i (g_i(x) > 0) + \sum_j (h_j(x) \neq 0)$$

- Quadratic penalty

$$p_{\text{quadratic}}(x) = \sum_i \max(g_i(x), 0)^2 + \sum_j h_j(x)^2$$

- Mixed Penalty

$$p_{\text{mixed}}(x) = p_{\text{count}} + p_{\text{quadratic}}$$

Barrier functions

$$p_{\text{barrier}}(x) = - \sum_i \log(-g_i(x))$$

Simplex problems

$$\begin{array}{ll} \min_x & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array}$$

multiple solutions

Inequality constraints

standard way of thinking using $\leq$

minimize problem such that something is less than

Equality constraints

Easier to solve using *s.l.e.* to solve for variables x_i

Conversion

Introduce *slacks*

$$a_i^T x \leq b \implies a_i^T x + s_i = b$$

The simplex algorithm

$$\mathcal{L}(x, \mu \geq 0, \lambda) = c^T x - \mu^T x - \lambda(Ax - b)$$

$$\nabla \mathcal{L} = c - \mu - \lambda^T A = 0 \implies A^T \lambda + \mu = c$$

Split into (B, N)

$$\begin{cases} A_B^T \lambda + \mu_B = c_B \\ A_N^T \lambda + \mu_N = c_N \end{cases} \implies \lambda = A_B^{-1} c_B \quad \text{since } x \geq 0, \mu_B = 0$$

Types

- Continuous
- Stochastic
- Deterministic